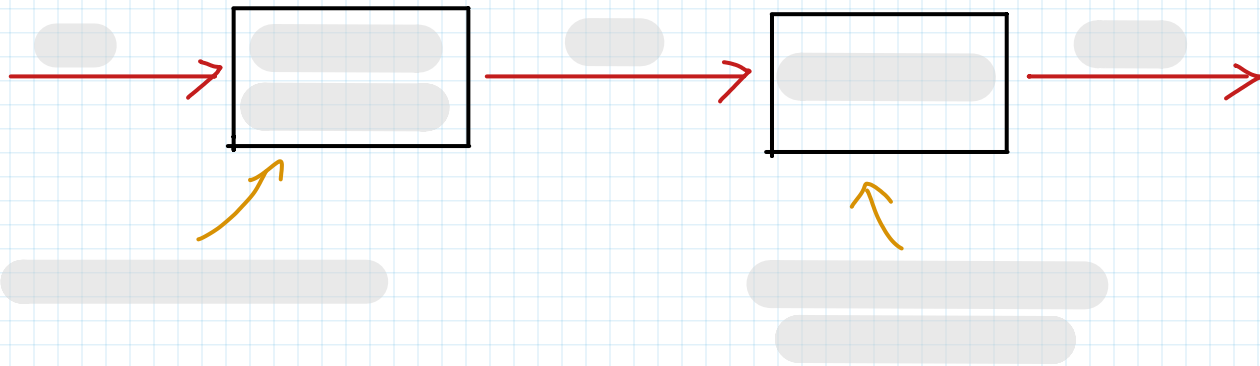


Where did we leave off?



r : Reference

u : Input

y : Output

$\Rightarrow$ To find a controller, we must first find out how the plant works!

$\Rightarrow$ Make a model of it!

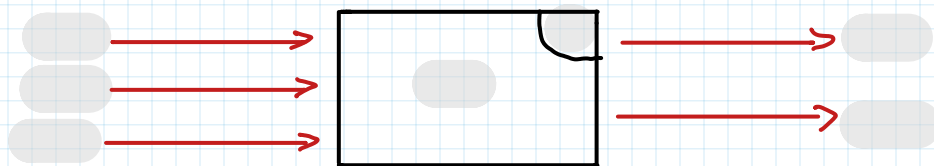
So how do we find this model?

$\Rightarrow$ Use equations to describe the dynamical behaviour of a system.

-system

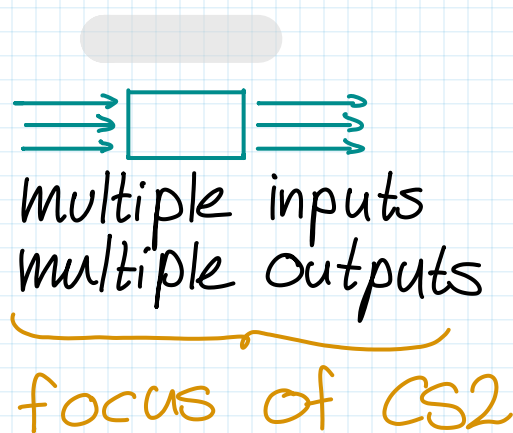
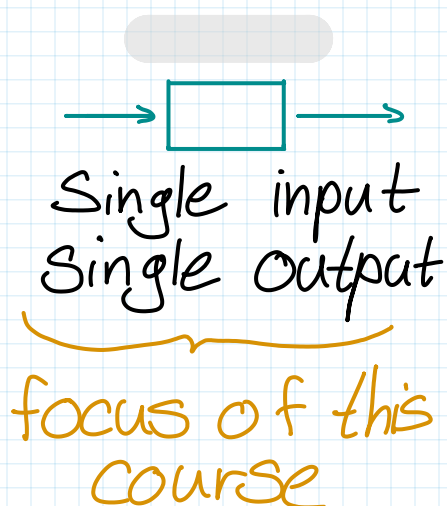
-system

The general system



- $u_i \hat{=}$ inputs
- $y_i \hat{=}$ outputs
- $\Sigma \hat{=}$ diff' eqn's of the system
- $x \hat{=}$

• Note: The general system can have multiple in-/outputs, and the #inputs can differ from the #outputs

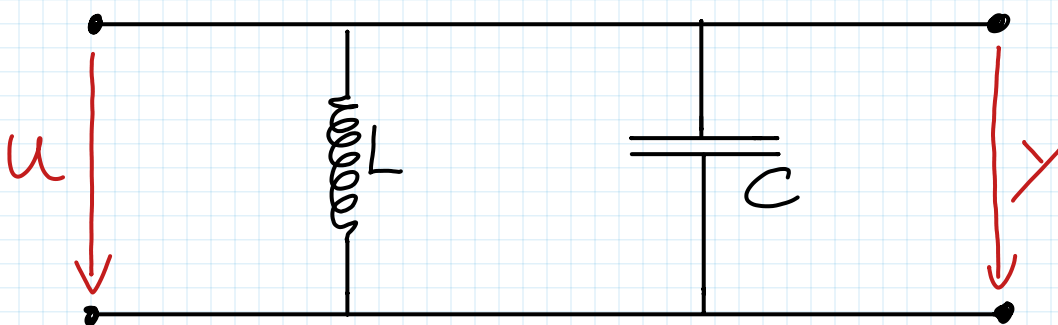


But what are state variables x ?

Def: Internal variable quantity. The output is influenced by the states. They are not always directly measurable if you only look at the input and output.

Example: RC - Circuit

- u : input voltage [V]
- y : output voltage [V]
- L : inductor [H]
- C : capacitor [F]



$\Rightarrow$ Output y is a function of the states at the capacitor and the states at the inductor (and the input u).

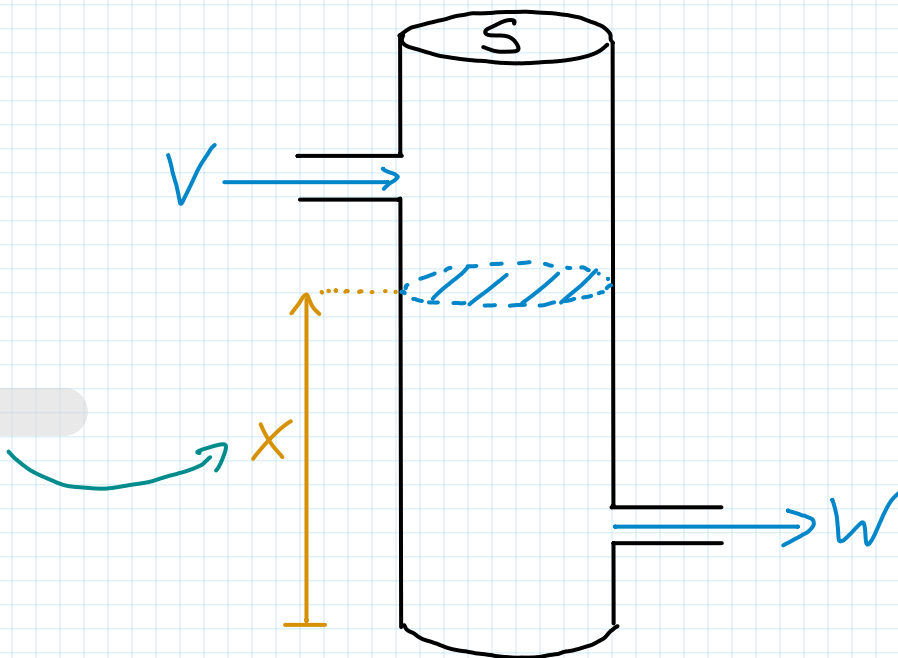
$$y = \underbrace{\quad\quad\quad}_{\quad\quad\quad}$$

- We do not see the states in the output, but the output is influenced by them!

- In many cases, the output is just one of the states, which simplifies the calculation.
- Let's look at an example from the lecture.

Example: Water tank

V : inflow [m^3/s]
 W : outflow [m^3/s]
 x : height [m]
 S : area [m^2]



- Now we can use a common modeling technique called the **naïve Bayes** based approach:

Think of the derivative as a " " " "

Here: Storage $\hat{=}$ water volume $\hat{=}$

- Simplifying:

$\Rightarrow$ Any model will take this form!

General form of diff' eqns:

	(I)
	(II)

(I): Internal system dynamics

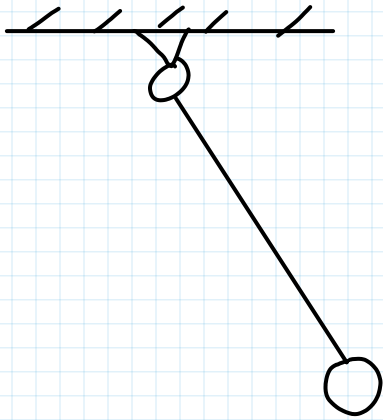
(II): System observation

Note: For a given system the eqns are not unique! We can measure multiple things on one system!
 Ex: Car output $\hat{=}$ ●, ●, ...

But the notation above only allows for first order derivatives... This can't describe any system right!

$\Rightarrow$ True! But theres a trick.

Example: pendulum



- $l, m, g \hat{=}$ constants
- $F_p \hat{=}$ Input
- $\theta \hat{=}$ Output

1) Describe dynamics
 $\Rightarrow$ Conservation of angular momentum

How do we
 achieve this
 form?

$$\begin{cases} \dot{x}(t) = f(x(t), u(t)) \\ y(t) = h(x(t), u(t)) \end{cases}$$

2) Introduce new state

• Rename: _____

• Define: _____

=> We deduce: _____

3) Write total system

• $\theta_1(t), \theta_2(t)$ are the _____ $x_i(t)$

• $F_p(t)$ is the _____ $u(t)$

• $y(t)$ is the _____ $y(t)$

4) Write in general form

$$\dot{x}_1(t) = x_2(t)$$

$$\dot{x}_2(t) = -\frac{g}{l} \sin(x_1(t)) + \frac{1}{ml} \cos(x_1(t)) u(t)$$

$$y(t) = x_1(t)$$

· We have achieved this form!

$$\begin{aligned}\dot{x}(t) &= f(x(t), u(t)) \\ y(t) &= h(x(t), u(t))\end{aligned}$$

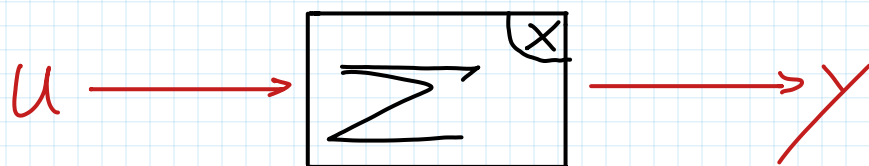
Intermediate Quiz questions

Q1) Conservation of angular momentum is a reservoir based modeling approach.
☐ TRUE ☐ FALSE

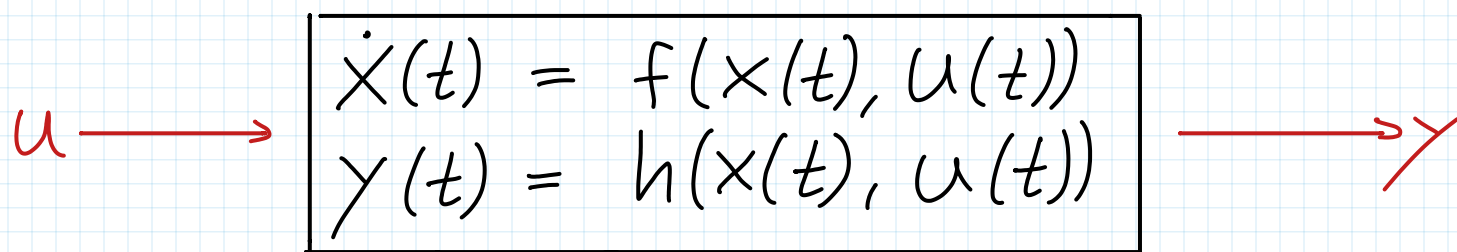
Q2) Inputs and Outputs are time dependent, but a general system is not.
☐ TRUE ☐ FALSE

Q3) The output of a system is independent of the chosen sensor.
☐ TRUE ☐ FALSE

How does this fit into the bigger picture?

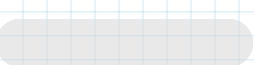


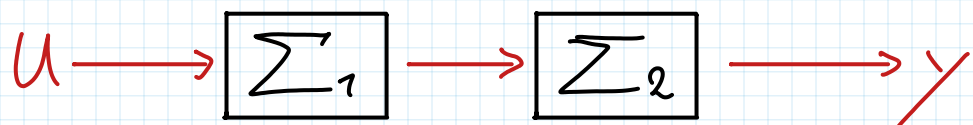
We now know what this truly is:



- Next week we will see what it means to **model** a system. Once we do that, we can write the signal-system relationship as such:

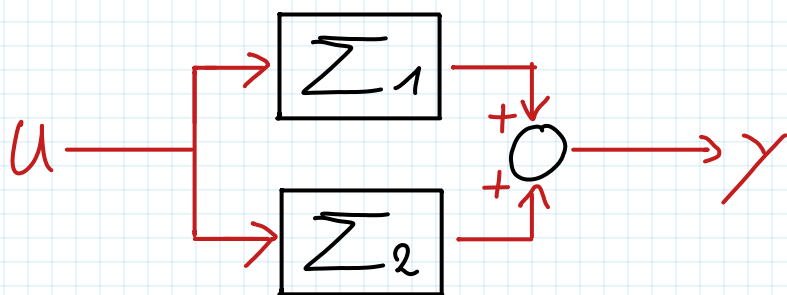
- This will allow us to analyze more complex interconnections of systems.
- In essence, any interconnection can be broken down in three fundamental connection types.

1)  interconnection

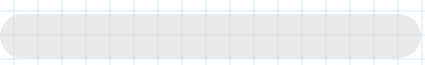


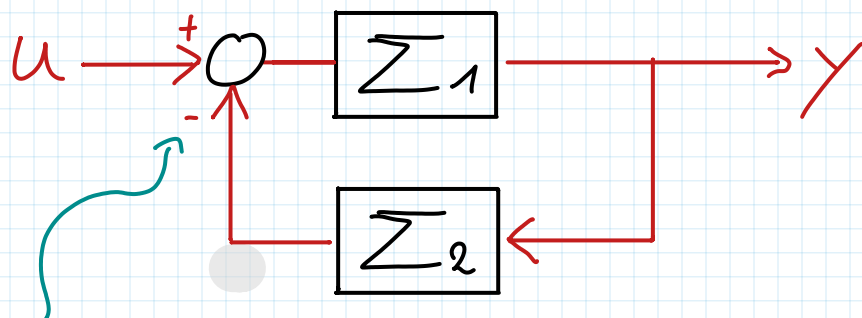
⇒

2) Parallel interconnection



⇒

3)  interconnection



Note: This defines if we have negative or positive feedback.

$\Rightarrow$

(I)

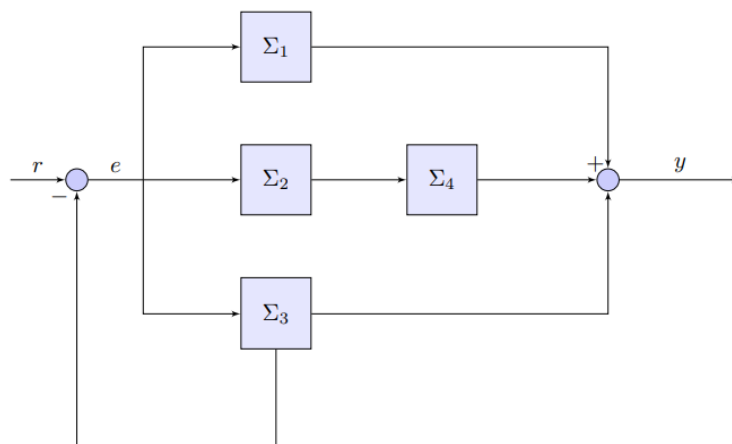
(II)

• Combining (I) and (II)

• Solve for y

• Exam 2018)

Question 5 Choose the correct answer. (1 Point)



You are given the above system diagram. What is the associated transfer function from $r \rightarrow y$.

☐ A $\Sigma_{r \rightarrow y} = (\Sigma_1 + \Sigma_2 \Sigma_4 + \Sigma_3)$

☐ B $\Sigma_{r \rightarrow y} = \frac{(\Sigma_1 + \Sigma_2 \Sigma_4 + \Sigma_3)}{1 + \Sigma_2 \Sigma_4}$

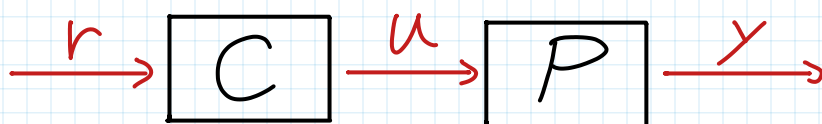
☐ C $\Sigma_{r \rightarrow y} = \frac{(\Sigma_1 + \Sigma_2 \Sigma_4 + \Sigma_3)}{1 + \Sigma_1 + \Sigma_2 \Sigma_4 + \Sigma_3}$

☒ $\Sigma_{r \rightarrow y} = \frac{(\Sigma_1 + \Sigma_2 \Sigma_4 + \Sigma_3)}{1 + \Sigma_3}$

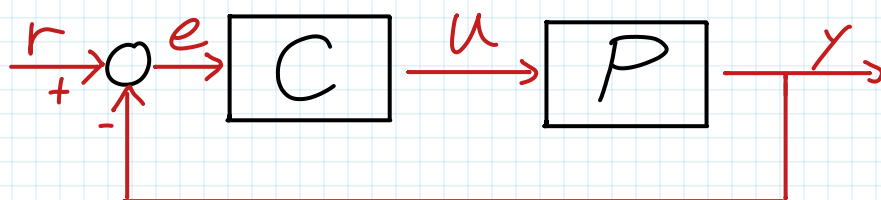
How do we now apply this to the structure we initially wanted to control?

• We had the following signals/systems:

- r : reference
- u : input
- y : Output
- C : Controller
- P : Plant



$\Rightarrow$



$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

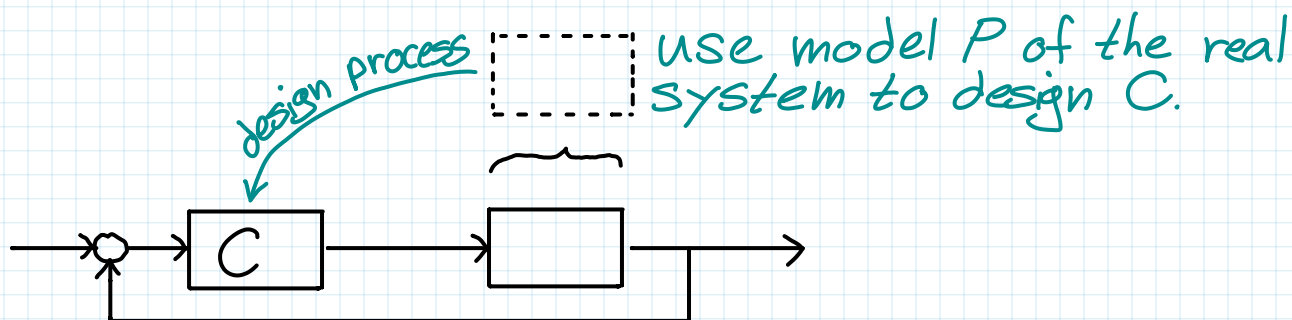
Why is feedback so crucial?

=> Feedforward only works if we have perfect knowledge of the plant in a perfect world.

- With feedback, we can:
 - + stabilize an unstable system
 - + compensate modeling uncertainties
 - + reject external disturbances
- But feedback can also:
 - introduce instability even in an otherwise stable system
 - feed noise into the system

Before we get to the quiz-questions, one final sidenote:

- What is the relation between Plant and model?



Q4) It is possible and a good idea to combine feedback and feedforward.

☐ TRUE ☐ FALSE

Q5) We treat system blockcharts just like resistances in a circuit.

☐ TRUE ☐ FALSE

Q6) When transforming a general system to a first order representation, we inevitably loose information.

☐ TRUE ☐ FALSE

Thanks for coming, have a wonderful weekend! ☺